

Full Length Article

A prompt tuning method based on relation graphs for few-shot relation extraction

Zirui Zhang^{a,*}, Yiyu Yang^b, Benhui Chen^{b,c}^a Nanjing University of Aeronautics and Astronautics, Nanjing, 210016, Jiangsu, China^b Dali University, Dali, 671000, Yunnan, China^c Lijiang Normal College, Lijiang, 674100, Yunnan, China

ARTICLE INFO

Keywords:

Relation extraction
Knowledge graph
Few-shot
Prompt tuning
Relation graph

ABSTRACT

Prompt-tuning has recently proven effective in addressing few-shot tasks. However, task resources remain severely limited in the specific domain of few-shot relation extraction. Despite its successes, prompt-tuning faces challenges distinguishing between similar relations, resulting in occasional prediction errors. Therefore, it is critical to extract maximum information from these scarce resources. This paper introduces the integration of global relation graphs and local relation subgraphs into the prompt-tuning framework to tackle this issue and fully exploit the available resources for differentiating between various relations. A global relation graph is initially constructed to enhance feature representations of samples across different relations based on label consistency. Subsequently, this global relation graph is partitioned to create local relation subgraphs for each relation type, optimizing the feature representations of samples within the same relation. This dual approach effectively utilizes the limited supervised information and improves tuning efficiency. Additionally, recognizing the substantial semantic knowledge embedded in relation labels, this study integrates such knowledge into the prompt-tuning process. Extensive experiments conducted on four low-resource datasets validate the efficacy of the proposed method, demonstrating significant performance improvements. Notably, the model also exhibits robust performance in discerning similar relations.

1. Introduction

Relation Extraction (RE) involves discerning relations between entities within unstructured text. The advent of pre-trained language models (PLMs) (Devlin, Chang, Lee, & Toutanova, 2018; Lewis et al., 2019; Liu et al., 2019; Raffel et al., 2020) has propelled the fine-tuning of PLMs as the primary methodology for relation extraction (Joshi et al., 2020; Xue, Sun, Zhang, & Chng, 2021; Zhou & Chen, 2021). Fig. 1 illustrates the fundamental concept of standard fine-tuning. However, fine-tuning necessitates augmenting PLMs with additional classifiers for further model training, making their performance heavily reliant on annotated data. Furthermore, the full potential of PLMs may not be realized due to discrepancies between the training objectives of PLMs and the specific requirements of downstream tasks.

Prompt-tuning, as proposed and validated in low-resource contexts (Gao, Fisch, & Chen, 2020; Scao & Rush, 2021), has emerged as an effective methodology (Brown et al., 2020; Liu et al., 2023;

Schick & Schütze, 2020a, 2020b). Illustrated in Fig. 1(b), the fundamental concept of prompt-tuning involves reshaping the objective of a downstream task to align more closely with that of a pre-trained task. This is achieved by designing a template to restructure input examples into well-formed phrases and linguistic constructs, facilitating label mapping to candidate words. The labels of input examples can then be inferred by predicting the masked tokens.

While prompt-tuning has shown promise in low-resource relation extraction tasks, it often overlooks the relations between labeled samples. To harness limited resources more effectively and discern differences between various relations, this study proposes the integration of global relation graphs and local relation subgraphs within the prompt-tuning framework. Initially, we enhance the feature representations of samples across different relations by constructing a global relation graph, where nodes represent labeled samples and edges signify similarity between samples. Subsequently, we refine the feature representations of samples within the same relation by constructing local

* Corresponding author.

E-mail addresses: zhangzirui@nuaa.edu.cn (Z. Zhang), yangyiyu@dali.edu.cn (Y. Yang), bhchen@dali.edu.cn (B. Chen).

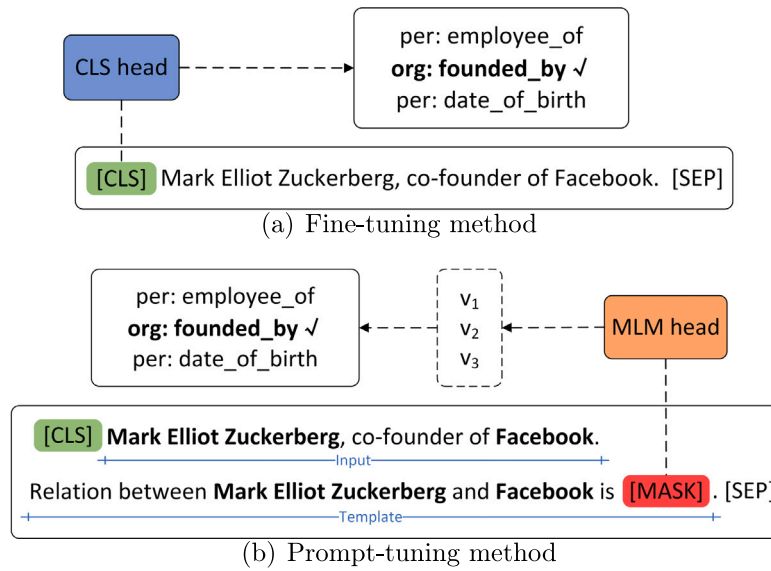


Fig. 1. Examples of fine-tuning and prompt-tuning core ideas.

relation subgraphs. Experimental results show that this approach improves the performance of relation extraction tasks in low-resource scenarios.

This paper conducts extensive experiments on four widely utilized datasets under low-resource conditions. By achieving superior performance compared to a range of recent baseline models, the efficacy of the proposed method is demonstrated. The contributions of this paper can be summarized as follows:

- (i) To effectively leverage the limited sample resources and capture the variability between different relations, we optimize the features of samples across various relations by constructing a global relation graph.
- (ii) Furthermore, to refine the feature representation of samples, local relation subgraphs are constructed to focus on samples within the same relation.
- (iii) The effectiveness of the proposed method in low-resource settings is validated through extensive experiments on four commonly used public datasets.

2. Related work

Early relation extraction was achieved by template matching rules generated by manual construction or automated mining. Its effectiveness relied heavily on the formulated rules, making it poorly generalized and challenging to migrate. With the development of deep learning, the performance of relation extraction tasks has been improved by leaps and bounds with the help of large amounts of labeled data. For example, Sui, Zeng, Chen, Liu, and Zhao (2023) constructed a relation extraction model based on a non-autoregressive parallel decoding Transformer network combined with an ensemble-based loss function. Existing deep learning-based relation extraction methods often require much-labeled data for the best results. However, high-quality labeled data is often complex to obtain in real-world scenarios, so these methods are usually difficult to apply directly.

To allow relation extraction models to have better generalization performance even when samples are scarce, many research efforts have turned to relation extraction in sample-less scenarios (Han et al., 2018; Xu et al., 2022). These approaches usually adopt the idea of prototype networks (Snell, Swersky, & Zemel, 2017). Gao, Han, Liu, and Sun (2019) designed instance-level and feature-level attention mechanisms based on prototype networks to highlight critical instances and features. Zhenzhen, Zhang, Nie, and Li (2022) learned prototype encoders from relation definition texts and used a joint training approach

to train relation definition prototype encoders and relation instance encoders. Li et al. (2023) proposed an entity with a Perceptual Embedding Module prototype attention network to learn meaningful entity representations and associations of entities with other words in the instance. These approaches typically combine word vector techniques with convolutional neural networks as encoders of textual semantics in relation contexts. Although CNNs can fuse global semantics and local critical information of relation contextual texts by convolution and pooling, their number of parameters is small. Therefore, it is difficult to capture the deep semantic features in the text, and the results of the relation extraction task with few samples are not satisfactory. After pre-training language models such as BERT and GPT were proposed, pre-training on unsupervised text and fine-tuning downstream tasks became a new paradigm for natural language processing tasks. As a result, some scholars have introduced pre-trained language models to the work of less-sample relation extraction. Feng, Wei, and Cui (2023) proposed a prototype network model for relation extraction based on entity convolution, where the head entity and tail entity embeddings of BERT encoding are transformed into several different convolutional kernels. Then, the convolution operation is performed on the original sentence to extract the features related to the entities in the sentence, and the prototype network is used to classify the extracted features. He, Niu, Feng, Wang, and Wei (2023) designed a multilevel embedding encoder with positional information and a Transformer to enhance the relation semantic representations by using the local information in the text. The encoded relation features are inputted into the prototype network for relation extraction.

The existing fine-tuning of pre-trained language models relies heavily on methods that add classifiers. A new approach to fine-tuning using pre-trained language models, prompt-tuning, has emerged (Ben-David, Oved, & Reichart, 2021; Lester, Al-Rfou, & Constant, 2021; Liu et al., 2023; Lu, Bartolo, Moore, Riedel, & Stenetorp, 2021; Reynolds & McDonnell, 2021). The method allows pre-trained language models to understand downstream tasks through prompt templates and label mapping. The prompt-tuning approach has a stronger generalization ability as it does not require adding additional classifiers to the model as in traditional fine-tuning methods. This allows the model to outperform traditional fine-tuning methods in relation extraction in fully supervised and sample-less scenarios (Lester et al., 2021). After the prompt learning approach was proposed, some works applied it to the relation extraction task with better results. Ye et al. (2022) took the ontology of the knowledge graph from outside of the task as the structural knowledge, converted it to text, and selected the information to be injected

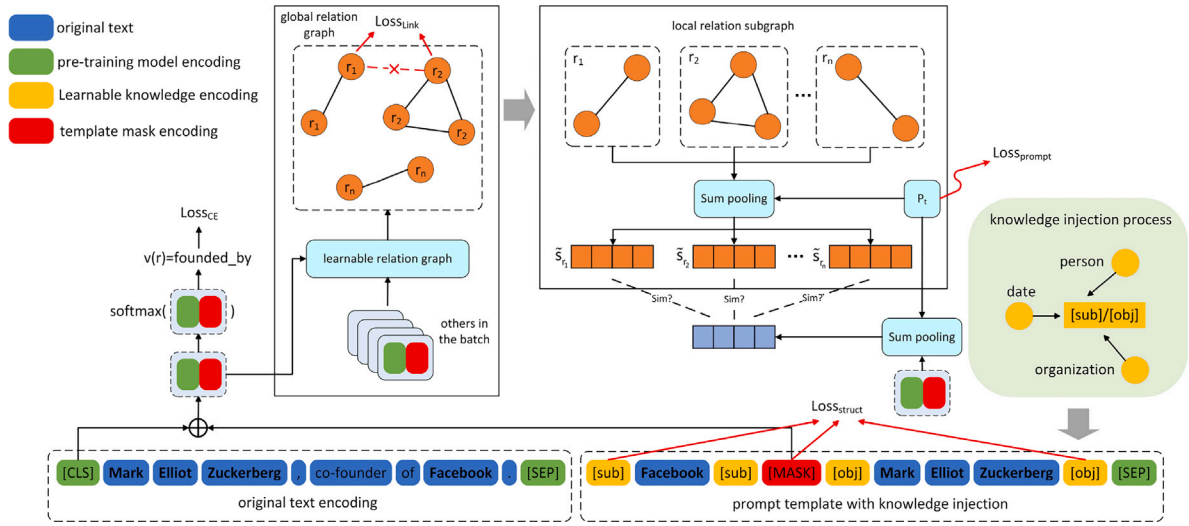


Fig. 2. Model architecture.

into the prompt templates using a visibility matrix. He, Huang, et al. (2023) proposed a dummy prompt pre-training method by mapping dummy prompts to semantic space with the pre-trained language model parameters fusion. The tag word mapping constructed by the prototype network is then used to model the mapping relation between entities and relations. PTR (Han, Zhao, Ding, Liu, & Sun, 2022) uses logic rules with multiple sub-prompts to encode prior knowledge in prompt-tuning. KnowPrompt (Chen et al., 2022) integrates knowledge from relation labels into prompt-tuning for relation extraction, facilitating co-optimization for improved performance. Despite the commendable performance of these methods, they often overlook the relations among a limited number of labeled samples. Consequently, misclassification can occur when discerning similar relations, ultimately leading to diminished performance metrics.

3. Background

The relation extraction task T comprises training, validation, and test datasets. Each sample instance x is mapped to the prompt input $x_{prompt} = T(\cdot)$ using a template. The template $T(\cdot)$ specifies the location and quantity of additional words to incorporate. While preserving the tokens in x , [MASK] is inserted into the prompt x_{prompt} for the language model to populate with tagged words. For example, consider a sentence from a relation extraction task:

$x_i = [CLS] \text{ text } [SEP]$ where the head and tail entities in the sentence are denoted as $Entity_1$ and $Entity_2$, respectively. Setting the template $T(\cdot) = "Entity_1 [MASK] Entity_2"$ maps x to:

$x_{prompt} = [CLS] \text{ text. } Entity_1 [MASK] Entity_2. [SEP]$ The tokens [CLS] and [SEP] are special token symbols used in language models. The vector corresponding to [MASK] is derived by transforming x into x_{prompt} prompts via language model encoding. A probability distribution $p([MASK]|x_{prompt})$ indicates which tokens are suitable for substituting [MASK] words. The linguistic function $v(\cdot)$ maps r_i to symbols representing the semantics of r_i .

Since the language model predicts the correct labeling of the masked position, we can formalize the masked position as $p(r|x)$, where:

$$p(r|x) = p([MASK] = v(r)|x_{prompt}) \quad (1)$$

The $\mathcal{L}_{CE}$ is calculated by the cross-entropy between r and $p(r|x)$ as shown below:

$$\mathcal{L}_{CE} = -\frac{1}{|\mathcal{X}|} \sum_{x \in \mathcal{X}} r \log p(r|x) \quad (2)$$

where $|\mathcal{X}|$ denotes the number of training data.

4. Methodology

This section illustrates the modeling diagram of a prompt-tuning method based on relation graphs for few-shot relation extraction. The specific model architecture is depicted in Fig. 2, and the subsequent sections elaborate on the construction and optimization of this model.

4.1. Knowledge injection module

Rich semantic knowledge in relation labeling is crucial. Drawing inspiration from the approach employed in KnowPrompt (Chen et al., 2022), this paper incorporates entity-related knowledge into constructing prompt templates.

4.1.1. Prompt template build with knowledge injection

Directly introducing type information for entities often necessitates additional annotation, which may not always be present in datasets. In this research, entity-type scopes are derived from prior knowledge about the relations. For example, with the relation “ $per : date_of_death$ ”, the subject entity is expected to be a “ $person$ ”, and the object entity is expected to be a “ $date$ ”. Therefore, prior distributions ϕ_{sub} and ϕ_{obj} can be estimated for the sets of possible entity type candidates C_{sub} and C_{obj} based on the relation classes. These prior distributions are calculated using frequency statistics. For instance, in a dataset with six relation labels, if subject entities of four relations are “ $organization$ ” and those of two relations are “ $person$ ”, then the prior distribution C_{sub} is computed as $\phi_{sub} = \{“organization” : 4/6, “person” : 2/6\}$. Initialization involves assigning type words around entities to aggregate embeddings with potential entity types, as illustrated below:

$$\hat{e}_{[sub]} = \sum \phi_{sub} \cdot e(I(C_{sub})) \quad (3)$$

$$\hat{e}_{[obj]} = \sum \phi_{obj} \cdot e(I(C_{obj})) \quad (4)$$

where $\hat{e}_{[sub]}$ indicates the typed word embedding associated with the subject entity, while $\hat{e}_{[obj]}$ denotes the typed word embedding linked to the object entity. The operation $I(\cdot)$ eliminates duplicate elements within a set. These type words provide an initial approximation of the possible entity types and encapsulate semantic information closely related to the proper entity types.

It is worth noting that the word embedding e comes from the pre-trained language model $\mathcal{L}$. The pre-trained language model chosen for this paper is BERT. As in the example given in the background, the template set up in this paper is $T(\cdot) = "Entity_1 [MASK] Entity_2"$. The input x is then mapped to $x_{prompt} = "[CLS] \text{ text. } Entity_1 [MASK] Entity_2. [SEP]"$.

[SEP]". Subsequently, the hidden vector corresponding to [MASK] is obtained by encoding x_{prompt} using the model $\mathcal{L}$. A probability distribution $p([MASK]|x_{prompt})$ is also generated, describing which tokens of the label words in the $\mathcal{L}$ vocabulary of the language model are suitable for replacing [MASK] words. The final encoded representation $\mathbf{e}$ is derived through $\mathcal{L}$.

4.1.2. Loss of knowledge injection module

In this study, structured constraints are employed to optimize prompts. Specifically, the ternary tuple (h, r, t) is used to describe relation facts, where h and t represent the types of subject and object entities, respectively, and r represents the relation label. The representation for h is derived from $\hat{\mathbf{e}}_{[sub]}$, while the representation for t is derived from $\hat{\mathbf{e}}_{[obj]}$. The representation for r is obtained from the relation labels encoded by a pre-trained model. The loss for implicit structured constraints, denoted as $\mathcal{L}_{struct}$, is defined as follows in this study:

$$\mathcal{L}_{struct} = -\log \sigma(\gamma - d_r(h, t)) - \sum_{i=1}^n \frac{1}{n} \log \sigma(d_r(h'_i, t'_i) - \gamma) \quad (5)$$

$$d_r(h, t) = \|h + r - t\|_2 \quad (6)$$

where (h'_i, r, t'_i) signifies a negative sample. The function σ is the sigmoid function, while γ represents the margin. The scoring function is denoted by d_r . This study assigns the accurate relation type to the [MASK] position during negative sampling. Negative samples are created by randomly selecting either subject or object entities to form irrelevant triples, ensuring that these entities belong to types that are impossible for the given relation.

4.2. Global relation graph module

In this section, the paper introduces the construction of a global relation graph to enhance the feature representation of samples across various relations. This approach enables the extraction of additional supervised signals from the training data by leveraging the global relation graph.

4.2.1. Construction of global relation graph

In a batch $B = \{(x_i, y_i)\}_{i=1}^M$ containing M labeled pairs, the indices are represented as $N = \{1, \dots, M\}$. A global relation graph is constructed from the sample data to enhance information extraction. This global relation graph is denoted by $G = \{V, E\}$, where V is the set of nodes and $E = \{e_{ij}\}$ is the set of edges between the samples. Each node $v_i \in V$ represents a sample x_i . An edge e_{ij} is established between nodes v_i and v_j if they share the same relation class. Specifically, e_{ij} is defined as follows:

$$e_{ij} = \begin{cases} 1 & y_j = y_i \\ 0 & \text{otherwise} \end{cases} \quad (7)$$

4.2.2. Loss of global relation graph module

The primary goal of creating a global relation graph G for a batch B is to connect nodes with the same relation while disconnecting those with different relations. The relation prediction $\hat{y}_i$ for node v_i , which corresponds to x_i , is calculated using equation (1).

In this study, $\hat{y}_i$ and $\hat{y}_j$ are established between v_i and v_j based on the correlation of $\hat{e}_{ij}$:

$$\hat{e}_{ij} = g(\hat{y}_i, \hat{y}_j) \quad (8)$$

where $\hat{y}_i$ and $\hat{y}_j$ are calculated from Eq. (1). $g(\cdot, \cdot)$ is the computed cosine similarity. The loss of $\mathcal{L}_{Link}$ is designed in this paper as:

$$\mathcal{L}_{Link} = - \sum_{i \in \mathcal{N}} \sum_{j \in \mathcal{C}(i)} e_{ij} \log(\hat{e}_{ij}) + (1 - e_{ij}) \log(1 - \hat{e}_{ij}) \quad (9)$$

where

$$\mathcal{C}(i) = \{j \in \mathcal{N} \text{ and } i \neq j\} \quad (10)$$

4.3. Local relation subgraph module

Drawing from constructing a global relation graph, this paper proceeds to enhance sample feature optimization within specific relations by creating a local relation subgraph.

4.3.1. Construction of local relation subgraph

Within a global relation graph G , consisting of a set of relation classes R and labeled nodes D , each relation $r \in R$ in the k -shot setting is defined by exactly k pairs of labeled nodes from D . A local relation subgraph is created for each relation $r \in R$. This local subgraph is transformed into an average representation of the current relation class. The vector $\tilde{\mathbf{S}}_r$ representing the current relation r is computed according to the following procedure:

$$\tilde{\mathbf{S}}_r = \frac{1}{k} \sum_{(v_i, y_i) \in D, y_i=r} \mathbf{p}_i \odot \mathbf{h}_{v_i} \quad (11)$$

where $\mathbf{h}_{v_i}$ denotes the feature vector of node v_i and $\mathbf{p}_i$ denotes the learnable prompt vector. The initialization of $\mathbf{p}_i$ is random. As a weight vector, its role is mainly to extract the most relevant prior knowledge by feature weighting the representation of the nodes.

After deriving the average representation $\tilde{\mathbf{S}}_r$ from the local relation subgraphs, the class label y_j for a node v_j outside the labeled set D is determined as follows:

$$y_j = \arg \max_{r \in R} \text{sim}(\mathbf{S}_{v_j}, \tilde{\mathbf{S}}_r) \quad (12)$$

A node is expected to be associated with the relation class that most closely resembles its local relation subgraph.

4.3.2. Loss of local relation subgraph modules

The approach adopted in this paper formulates the loss function based on subgraph similarity to optimize the learnable prompt $\mathbf{p}_i$. The prompt tuning loss $\mathcal{L}_{prompt}$ on the labeled training set D is defined as:

$$\mathcal{L}_{prompt}(\mathbf{p}_i) = - \sum_{(x_i, y_i) \in D} \ln \frac{\exp(\text{sim}(\mathbf{S}_{x_i}, \tilde{\mathbf{S}}_{y_i})/k)}{\sum_{r \in R} \exp(\text{sim}(\mathbf{S}_{x_i}, \tilde{\mathbf{S}}_r)/k)} \quad (13)$$

where

$$\mathbf{S}_{x_i} = \mathbf{p}_i \odot \mathbf{h}_{v_i} \quad (14)$$

where the aggregation feature $\tilde{\mathbf{S}}_r$ represents the local relation subgraph for relation r . $\mathbf{S}_{x_i}$ denotes the aggregation feature of node v_i .

4.4. Joint training

This paper introduces four distinct types of losses for joint training: the cross-entropy loss $\mathcal{L}_{CE}$, the structuring loss $\mathcal{L}_{struct}$, the link prediction loss $\mathcal{L}_{Link}$, and the learnable prompts loss $\mathcal{L}_{prompt}$. Each loss is computed using specific Eqs. (2), (5), (9), and (13) respectively. The total loss $Loss$ is determined as:

$$Loss = \mathcal{L}_{CE} + \alpha \mathcal{L}_{struct} + \beta \mathcal{L}_{Link} + \gamma \mathcal{L}_{prompt} \quad (15)$$

where α , β and γ are the weight parameters between losses.

5. Experiment

This paper assesses the efficacy of the proposed methodology across four datasets. Comparative analyses are conducted against state-of-the-art baseline models, followed by additional experiments and analyses focusing on the fundamental components of the model.

5.1. Dataset

In this paper, experiments are conducted on four commonly used datasets: SemEval (Hendrickx et al., 2019), TACRED (Zhang, Zhong, Chen, Angeli, & Manning, 2017), TACREV (Alt, Gabryszak, & Hennig, 2020), and Re-TACRED (Stoica, Platanios, & Póczos, 2021). Table 1 provides detailed information for each dataset.

Table 1
Statistical results of dataset information.

Dataset	Train	Val	Test	Rel
SemEval	6,507	1,493	2,717	19
TACRED	68,124	22,631	15,509	42
TACREV	68,124	22,631	15,509	42
Re-TACRED	58,465	19,584	13,418	40

Table 2
Experimental results for standard conditions.

Model	SemEval	TACRED	TACREV	Re-TACRED
SpanBERT (Joshi et al., 2020)	–	70.8	78.0	85.3
KnowBERT (Peters et al., 2019)	89.1	71.5	79.3	89.1
LUKE (Yamada, Asai, Shindo, Takeda, & Matsumoto, 2020)	–	72.7	80.6	–
MTB (Soares, Fitzgerald, Ling, & Kwiatkowski, 2019)	89.5	70.1	–	–
GDPNet (Xue et al., 2021)	–	71.5	79.3	–
PTR (Han et al., 2022)	89.9	72.4	81.4	90.9
KnowPrompt (Chen et al., 2022)	90.2	72.4	82.4	91.3
ours	90.6	73.2	80.7	89.6

“–” indicates metrics not reported in the paper.

5.2. Experimental setup

This paper conducts two experiments to verify the method’s validity: one in a standard scenario and the other in a low-resource scenario. In the standard scenario, the full dataset D_{train} is used for training. For the low-resource scenario, three different sample sizes are set for training, specifically $K = 8$, $K = 16$, and $K = 32$. This paper uses the $F1$ value as the final evaluation metric. The experimental results demonstrate that the proposed method performs better in standard and low-resource scenarios.

5.3. Baseline comparison

Table 2 lists the experimental results of the various models on the four datasets under the standard scenario. From the experimental results, it can be seen that although the method focuses on the problem of dealing with few-shot samples, it still performs well under standard settings.

Tables 3 and 4 present the experimental results of the various models on the four datasets in the low-resource scenario. This paper

categorizes all models into two groups: fine-tuning-based and prompt-based. Each dataset is configured in three different low-resource scenarios for separate experiments, with sample sizes set to $K = 8$, $K = 16$, and $K = 32$. The results demonstrate the superior performance of the proposed method.

From the experimental results, it can be seen that the prompt-based approach is generally better than the fine-tuning-based approach. Especially in very low-resource scenarios, prompt-based methods show excellent performance. The model proposed in this paper outperforms the state-of-the-art methods, exceeding baseline models in fine-tuning and prompt-based approaches. This shows that the core module of this method is indeed feasible. Furthermore, it demonstrates robust performance across all four datasets, particularly in low-resource scenarios.

5.4. Ablation study

In this paper, we conduct ablation studies on two publicly available datasets, systematically removing one component at a time to analyze the impact on model performance. The results of these ablation studies are summarized in Table 5.

Four types of removal studies were conducted in Table 5, including removing the global relation graph module, the local relation subgraph module, the knowledge injection module, and the negative sample. The results show that removing these core components results in varying degrees of model performance degradation, highlighting the effectiveness of each component in improving model performance. This suggests that the injection of semantic knowledge and the construction of both global relation graphs and local relation subgraphs have important roles.

5.5. Feature visualization and analysis

To provide a more intuitive understanding of the changes in feature distribution, this paper employs t-SNE on the Re-TACRED dataset to map the feature distribution. Each set of experiments was visualized by randomly selecting a batch of samples. Firstly, feature visualization is conducted at various stages of the model to verify the effectiveness of its components. Secondly, a comparison is made with KnowPrompt, a method known for its robust performance.

5.5.1. Visualization of features at each stage of the model

To validate the efficacy of each component in the model, feature visualization is performed at different stages. The distribution of features at each stage is illustrated in Fig. 3. Each of the three figures represents the results of the feature visualization after the introduction of the new module on top of the previous figure.

As depicted in Fig. 3(a), introducing knowledge injection based on prompt-tuning results in a more pronounced aggregation phenomenon among features. However, there remains a close distance between

Table 3
Experimental results on datasets TACRED and TACREV in the low resource conditions.

Model	TACRED			TACREV		
	K = 8	K = 16	K = 32	K = 8	K = 16	K = 32
Fine-tuning methods						
SpanBERT (Joshi et al., 2020)	8.4	17.5	17.9	5.2	5.7	18.6
LUKE (Yamada et al., 2020)	9.5	21.5	28.7	9.8	22.0	29.3
GDPNet (Xue et al., 2021)	11.8	22.5	28.8	8.3	20.8	28.1
TANL (Paolini et al., 2021)	18.1	27.6	32.1	18.6	28.8	32.2
Prompt-tuning methods						
TYP Marker(punct)(Zhou & Chen, 2021)	28.9	32.0	32.4	27.6	31.2	32.0
ENT Marker(punct)(Zhou & Chen, 2021)	27.0	31.3	31.9	27.4	31.2	32.0
AdaPrompt (Chen et al., 2021)	–	–	–	25.2	27.3	30.8
PTR (Han et al., 2022)	28.1	30.7	32.1	28.7	31.4	32.4
KnowPrompt (Chen et al., 2022)	32.0	35.4	36.5	32.1	33.1	34.7
ours	34.1	37.2	40.3	34.9	37.4	39.2

“–” indicates metrics not reported in the paper.

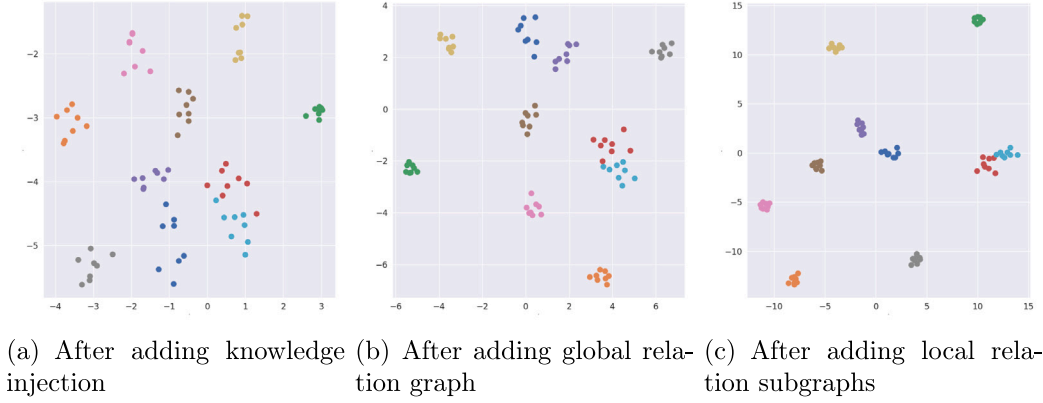


Fig. 3. Results of the visualization of the features at each stage.

Table 4

Experimental results on datasets Re-TACRED and SemEval in the low resource conditions.

Model	Re-TACRED			SemEval		
	K = 8	K = 16	K = 32	K = 8	K = 16	K=32
Fine-tuning methods						
SpanBERT (Joshi et al., 2020)	14.2	29.3	43.9	–	–	–
LUKE (Yamada et al., 2020)	14.1	37.5	52.3	–	–	–
GDPNet (Xue et al., 2021)	18.8	48.0	54.8	42.0	67.5	81.2
TANL (Paolini et al., 2021)	26.7	50.4	59.2	–	–	–
Prompt-tuning methods						
TYP Marker(punct)(Zhou & Chen, 2021)	44.8	54.1	60.0	–	–	–
ENT Marker(punct)(Zhou & Chen, 2021)	44.1	53.7	59.9	–	–	–
PTR (Han et al., 2022)	51.5	56.2	62.1	70.5	81.3	84.2
KnowPrompt (Chen et al., 2022)	55.3	63.3	65.0	74.3	82.9	84.8
ours	59.7	65.6	67.3	79.1	83.2	84.6

“–” indicates metrics not reported in the paper.

Table 5

Ablation study results.

Model	TACRED			TACREV		
	K = 8	K = 16	K = 32	K = 8	K = 16	K=32
ours	34.1	37.2	40.3	34.9	37.4	39.2
-global relation graph	31.9	36.0	36.2	32.8	35.1	36.6
-local relation subgraph	32.2	36.1	36.4	32.2	35.4	36.7
-knowledge injection	32.4	35.8	35.3	32.4	35.5	37.9
-negative sample	33.8	37.0	38.6	33.7	37.2	38.8

different relations, and the distribution of features within the same relation appears loose, potentially leading to misclassification during final judgment. Fig. 3(b) illustrates that the distance between different relations becomes more distinct upon introducing the global relation graph. This enhancement suggests improved separation of relation-specific features. Furthermore, incorporating local relation subgraphs, as demonstrated in Fig. 3(c), leads to a more clustered distribution of features within the same relation. This indicates a finer distinction and better representation of relation-specific characteristics. The t-SNE visualization indirectly confirms the effectiveness of the proposed model in enhancing feature separation and clustering, which are critical for accurate relation extraction.

5.5.2. Comparison of model visualizations under different conditions

This paper conducts a detailed comparison with the robust baseline model KnowPrompt. Feature visualization results of the two methods are presented under settings with $K = 8$, $K = 16$, and $K = 32$ for three cases. The visualization outcomes of the two methods under different settings are depicted in Fig. 4.

The results indicate that our method demonstrates superior aggregation compared to the KnowPrompt method for $K = 8$ and $K = 16$.

This improvement can be attributed to the constraints imposed by the global relation graph and local relation subgraph, which optimize the feature representation. However, at $K = 32$, both methods exhibit similar performance, as they have adequate sample sizes to learn a more refined feature representation.

5.6. Detailed analysis

This section will further analyze the methodology’s effectiveness, using the Re-TACRED dataset as an example. Firstly, various types of relations in the dataset are compared and analyzed. Secondly, specific cases are compared and analyzed. Finally, similar relations are visualized and analyzed. The validity of the method is verified through multiple perspectives.

5.6.1. Comparative analysis of relations

In this paper, we compare and analyze the strong baseline model KnowPrompt on all kinds of relations, and the specific experimental results are shown in Fig. 5. The specific meanings of each type of relation are in Table A1 in the Appendix. From the experimental results in Fig. 5, it can be seen that among the 37 types of relations present in the dataset Re-TACRED, the accuracy of this method is higher than or equal to the baseline model KnowPrompt for 30 types of relations, which indicates that this method is effective in predicting most of the relations.

It is worth noting that for some relation types that were not predicted very well by the baseline model, the predictions of the present model achieved a hit rate of 100%. For example, the “org : number_of_employees/members” relation is represented by R_{10} and the “per : stateorprovince_of_death” relation is represented by R_{11} . In addition, the hit rate of this method is higher than that of the strong

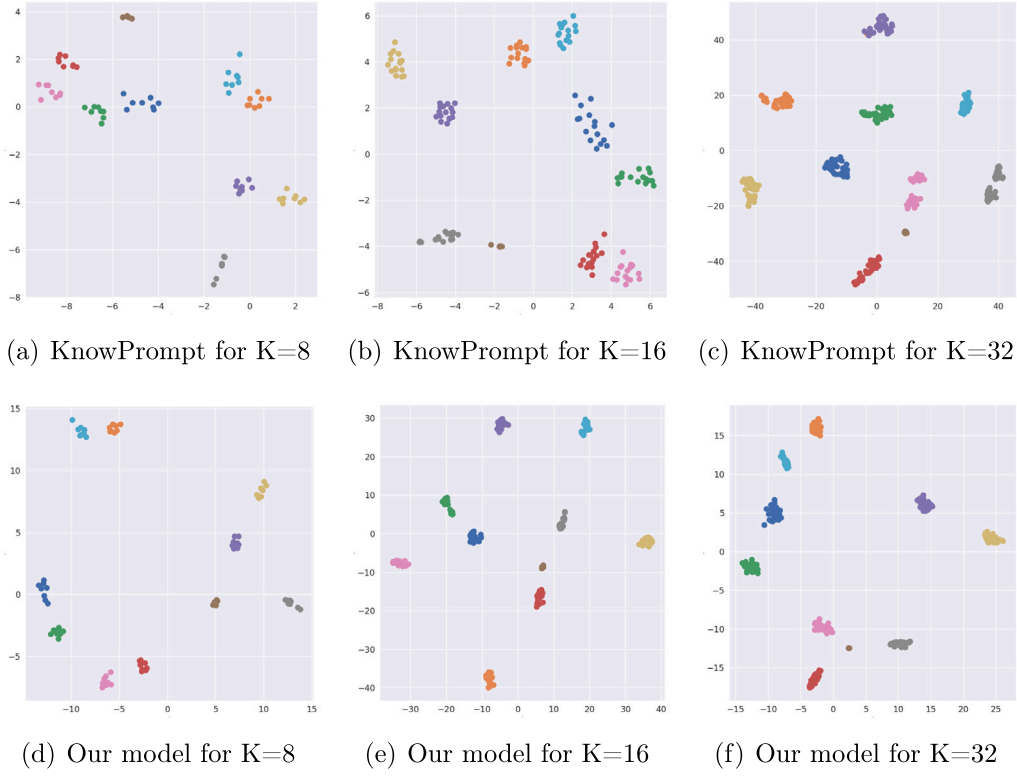


Fig. 4. Results of feature visualization comparison between our model and KnowPrompt.

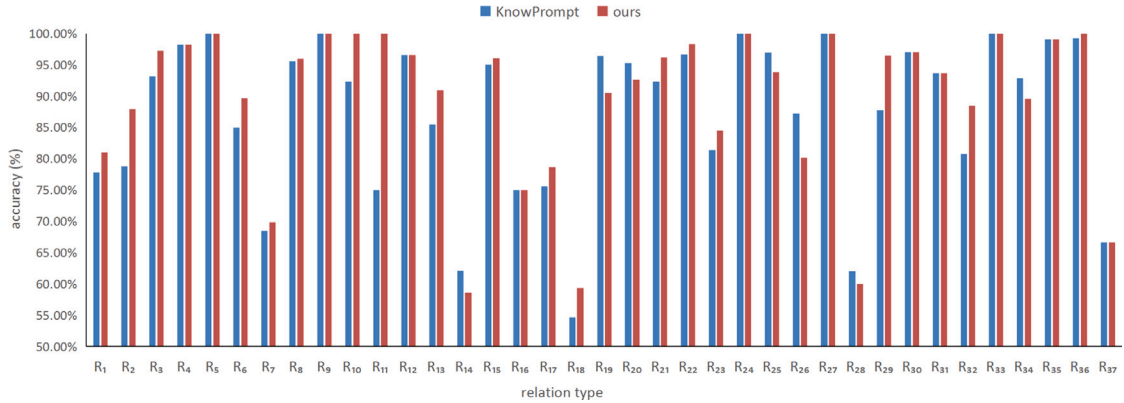


Fig. 5. Comparison results of various types of relation predictions in the dataset Re-TACRED.

baseline model for more similar relations. For example, the “*per : spouse*” relation is represented by R_3 and the “*per : parents*” relation is represented by R_6 .

5.6.2. Case study

A case study is presented in this section to provide insight into the methodology and demonstrate its effectiveness. Table 6 shows the results of the case study compared to a strong baseline. The methodology can better differentiate between difficult tasks. In the validation set, all relations can be classified into two categories: “*person*” and “*organization*”. This paper selects several samples with similar relations from these two categories to analyze their classification results. Seven samples are classified on different models, as shown in Table 6, and the proposed model obtains correct classification results in these samples. For example, in the second sample, the model gives correct results, while the models PTR and KnowPrompt give wrong results. This example is difficult for the model because the sentence contains

the interference “*husband*”. An important reason for the model to be able to differentiate may be the introduction of a global relation graph, which distances the different relations and optimizes the feature representation of the samples. Therefore, the components of the present method are effective.

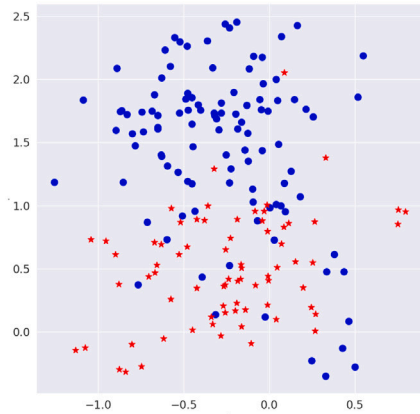
5.6.3. Visual analysis of similarity relations

In this paper, we argue that one of the main reasons for the failure of classification prediction is the existence of some hard-to-distinguish relations, especially those relation classes with high similarity. Therefore, t-SNE provides an intuitive visualization of the feature representation for similar relations. The specific visualization results are shown in Fig. 6. Specifically, two similar relations R_3 (“*per : spouse*”) and R_6 (“*per : parents*”) are selected for analysis in this paper, and several instances are randomly selected from the two relations that were visualized. As can be seen from the visualization results in Fig. 6, the feature embeddings of the similar relations are relatively separate

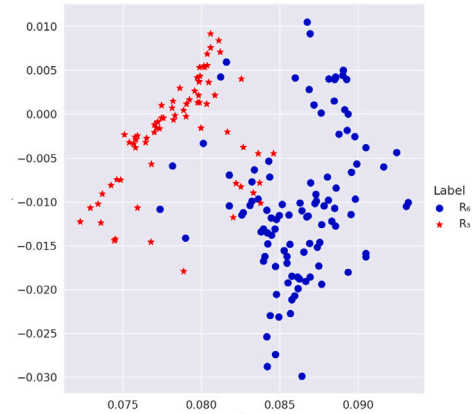
Table 6

A case study of modeling on similar relations. In each sentence, the head entity is highlighted in blue, and the tail entity is highlighted in yellow.

Similar relation	Sentence	PTR	KnowPrompt	Ours
per: per:identity, per:children, per:spouse, etc	Xie was the sister-in-law of Wen Qiang, formerly the second in command of the Chongqing police and director of the justice bureau before he was arrested.	per:siblings (R_2) ✗	per:identity (R_{23}) ✗	per:other_family (R_{21}) ✓
	She played the joke on her husband, David.	per:children (R_{13}) ✗	per:spouse (R_3) ✗	per:identity (R_{23}) ✓
	Here is the proof: Nir Rosen resigns after remarks about CBS reporter Lara Logan Rosen, a fellow reporter, has been forced to resign his position at NYU over comments he made regarding Logan.	per:schools _attended (R_{30}) ✗	per:schools _attended (R_{30}) ✗	per:employee_of (R_{17}) ✓
	A man who shot and killed three women in a Pennsylvania health club, then himself, apparently blogged his rage-filled preparations –with the final chilling entry announcing the “big day” of the massacre.	per:stateorprovince _of_death (R_{11}) ✓	per:stateorprovinces _of_residence (R_7) ✗	per:stateorprovince _of_death (R_{11}) ✓
org: org:member_of, org:shareholders, org:members	AIG said Monday it had also raised 16.2 billion dollars by selling unit American Life Insurance Company ALICO to MetLife Inc.	org:shareholders (R_{16}) ✗	org:shareholders (R_{16}) ✗	org:member_of (R_{18}) ✓
	The first \$9 billion in proceeds from the sale will go toward redeeming preferred shares in ALICO held by the Federal Reserve Bank of New York , with the remainder to be used to reduce the New York Fed’s lending commitment to AIG.	org:member_of (R_{18}) ✗	org:shareholders (R_{16}) ✓	org:shareholders (R_{16}) ✓
	PDA, The Progressive Democrats of America works creatively inside and outside the Democratic Party .	org:members (R_1) ✗	org:member_of (R_{18}) ✓	org:member_of (R_{18}) ✓



(a) Before introducing this method



(b) After introducing this method

Fig. 6. Feature visualization results for the similarity relations R_3 (“per : spouse”) and R_6 (“per : parents”) before and after the introduction of the present method.

after introducing the present method, making it easier to classify them. This experiment can prove that the present model can produce better discriminative features, especially for challenging tasks.

6. Conclusion

This paper introduces a prompt tuning method based on relation graphs for few-shot relation extraction. The method constructs a global relation graph by leveraging labeling consistency among samples to optimize feature representations across different relations. Subsequently, the global relation graph is partitioned into local relation subgraphs

to refine feature representations within the same relation. This approach maximizes the utilization of limited supervised information and enhances tuning efficiency. Experimental validation on four publicly available datasets demonstrates the effectiveness of the proposed method, showing superior performance compared to existing methods in low-resource scenarios.

CRedit authorship contribution statement

Zirui Zhang: Writing – original draft, Methodology. **Yiyu Yang:** Writing – review & editing. **Benhui Chen:** Supervision.

Table A.1

Correspondence table between relation id and specific relation types.

Relation id	Relation type	Relation id	Relation type
R_1	org:members	R_{20}	per:countries_of_residence
R_2	per:siblings	R_{21}	per:other_family
R_3	per:spouse	R_{22}	per:religion
R_4	org:country_of_branch	R_{23}	per:identity
R_5	per:country_of_death	R_{24}	per:date_of_birth
R_6	per:parents	R_{25}	org:city_of_branch
R_7	per:stateorprovinces_of_residence	R_{26}	org:alternate_names
R_8	org:top_members/employees	R_{27}	org:website
R_9	org:dissolved	R_{28}	per:cause_of_death
R_{10}	org:number_of_employees/members	R_{29}	org:stateorprovince_of_branch
R_{11}	per:stateorprovince_of_death	R_{30}	per:schools_attended
R_{12}	per:origin	R_{31}	per:date_of_death
R_{13}	per:children	R_{32}	per:city_of_death
R_{14}	org:political/religious_affiliation	R_{33}	org:founded
R_{15}	per:title	R_{34}	per:cities_of_residence
R_{16}	org:shareholders	R_{35}	per:age
R_{17}	per:employee_of	R_{36}	per:charges
R_{18}	org:member_of	R_{37}	per:stateorprovince_of_birth
R_{19}	org:founded_by		

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

See Table A.1

Data availability

Data will be made available on request.

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